

Name _____ Date _____ Period _____

Parent Functions IV: b^x and log base b

Algebra II | Unit 1 | Day 7 | TEK A2.2.A

Objective: I can graph exponential and logarithmic parent functions for $b = 2, 10$, and e , analyze their key attributes, and notice that their domains and ranges trade places.

Part 1 — The Exponential Function, $f(x) = 2^x$

Fill in the table. Negative exponents give fractions, never negative outputs.

x	-2	-1	0	1	2	3
$f(x) = 2^x$						

Can 2^x ever equal 0 or a negative number? _____

So there is a horizontal asymptote at _____ and a y-intercept at _____.

Part 2 — The Logarithmic Function, $f(x) = \log_2(x)$

This undoes the exponential. Compare it to your Part 1 table.

x	1/4	1/2	1	2	4	8
$f(x) = \log_2(x)$						

What happened to the table from Part 1? _____

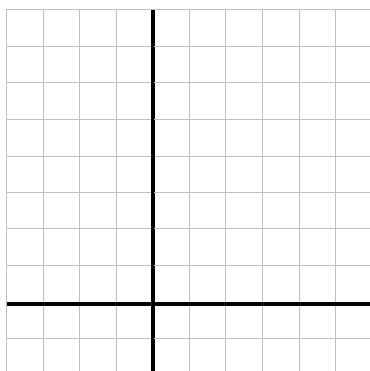
What is $\log_2(0)$? _____

So there is a vertical asymptote at _____ and an x-intercept at _____.

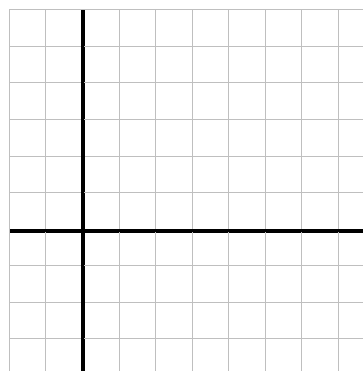
Part 3 — Graph Them Both

Each square is 1 unit. Sketch the asymptote as a dashed line first, then plot.

A. $f(x) = 2^x$



B. $f(x) = \log_2(x)$



Asymptote: _____

Asymptote: _____

Part 4 — Compare the Key Attributes

Attribute	$f(x) = b^x$	$f(x) = \log \text{ base } b$
Domain (interval)	_____	_____

Range (interval)	_____	_____
Intercept	_____	_____
Asymptote	_____	_____
Increasing / decreasing	_____	_____

Look at the first two rows. The domain and range have _____, and so did the intercepts and asymptotes. That is a strong hint these are _____ - Day 10.

Part 5 — Changing the Base

All bases greater than 1 give the same shape. Only the steepness changes.

x	0	1	2
2^x	1	2	4

x	0	1	2
10^x	1	10	100

Which grows faster, and why? _____

What do all three exponential graphs share? _____

Practice

Show your work where there is room.

- From your Part 1 table, a negative exponent on 2 always gives a _____, never a negative number.
- Evaluate: $\log_2(8) = \underline{\hspace{2cm}}$ (because $2^3 = 8$)
- Solve $2^x = 16$. $x = \underline{\hspace{2cm}}$
- Solve $\log_2(x) = 5$. $x = \underline{\hspace{2cm}}$
- A function has domain $(0, \infty)$, range all reals, and passes through $(1, 0)$. Name it. _____
- A function has a horizontal asymptote $y = 0$ and y-intercept $(0, 1)$. Name it. _____
- True or false: $\log_2(0)$ is defined. _____
- Explain why $\log_2(0)$ is undefined - what power of 2 could possibly give 0?

Exit Ticket

A function has domain $(0, \infty)$, range $(-\infty, \infty)$, an x-intercept at $(1, 0)$, and a vertical asymptote at $x = 0$.

Name the family: _____ Name its inverse family: _____